P P SAVANI UNIVERSITY

BCA (Hons.) Semester VII Examination

October 2026

SSCA4020 Full Stack Development

14.10.2026, Wednesday **Time:12:30 pm to 03:00 pm** **Maximum Marks: 60**

Instructions:

1. The question paper comprises of two sections.
2. Answer questions as per the internal choices given.
3. Assume suitable data wherever necessary and write neat, well-structured code where required.

SECTION – I

Q - 1	**MCQ/Short Question/Fill in the Blanks (Any Five)**	**[05]**	**CO**	**BTL**
(i)	Which HTML5 element is used to define navigation links? A. <header> B. <nav> C. <section> D. <footer>		CO3	1
(ii)	What is the main purpose of CSS? A. To store data in a database B. To create server-side logic C. To style and format web pages D. To define database tables		CO3	1
(iii)	Define client-server architecture.		CO1	2
(iv)	The _____ in JavaScript represents an HTML document as a tree of objects.		CO4	1
(v)	Name any two JavaScript data types.		CO4	1
(vi)	What is responsive web design?		CO2	2
(vii)	State one advantage of using Tailwind CSS.		CO2	2
Q - 2	**Attempt any two.**	**[10]**		
(i)	Explain client-side and server-side development with one example each.		CO1	2
(ii)	Describe semantic HTML elements and give any four examples.		CO3	2
(iii)	Explain the CSS box model with a neat diagram.		CO3	2
(iv)	Write a JavaScript function to check whether a number is even or odd.		CO4	3
Q - 3	**Attempt any two.**	**[10]**		
(i)	Explain let, const and var in JavaScript.		CO4	2
(ii)	What are JavaScript events? Explain any two events with examples.		CO4	2
(iii)	Explain jQuery selectors and demonstrate any two selectors with syntax.		CO4	2
Q - 4	**Attempt any one.**	**[05]**		
(i)	Explain the Agile development workflow and the role of Git in a team project.		CO1	2
(ii)	Explain Tailwind CSS utility classes and how they help build a responsive layout.		CO2	2

SECTION – II

Q - 1	**MCQ/Short Question/Fill in the Blanks (Any Five)**	**[05]**	**CO**	**BTL**
(i)	Which command is commonly used to start a Node.js application? A. npm install B. node app.js C. git start D. npm build		CO5	1
(ii)	What is Express.js? A. A database management system B. A CSS framework C. A web framework for Node.js D. A JavaScript data type		CO5	1
(iii)	Define RESTful API.		CO5	1

(iv)	What is the purpose of a React component?		CO2	1
(v)	What is JSX in React?		CO2	1
(vi)	_____ is documentoriented NoSQL database used to store data in flexible,JSON-like documents.		CO5	1
(vii)	What is JWT used for?		CO5	1
Q - 2	**Attempt any two.**	**[10]**		
(i)	Explain props and state in React with a suitable example.		CO2	2
(ii)	Describe the role of React Router in a single-page application.		CO2	2
(iii)	Explain middleware in Express.js and mention one use case.		CO5	2
(iv)	Differentiate SQL and NoSQL databases with examples.		CO5	2
Q - 3	**Attempt any two.**	**[10]**		
(i)	Explain the role of Node.js and Express.js in backend web development.		CO5	3
(ii)	Explain CRUD operations in MongoDB with suitable examples.		CO5	2
(iii)	Explain password hashing and the role of bcrypt in authentication.		CO5	2
Q - 4	**Attempt any one.**	**[05]**		
(i)	Explain the basic steps involved in deploying a web application using any one platform mentioned in the syllabus.		CO5	2
(ii)	Describe how a React frontend can communicate with a Node.js/Express backend using a REST API.		CO5	2

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CO : Course Outcome Number BTL : Blooms Taxonomy Level

Level of Bloom's Revised Taxonomy in Assessment

1: Remember	2: Understand	3: Apply
4: Analyze	5: Evaluate	6: Create

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NAAC ACCREDITED A+ GRADE